From Detection to Explanation: Naming the Device, and the Limits of Instructions

Technical report on the explanation layer of the contrastive-pairs diagnostic (v0.3, 60 pairs)

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1. Goal

Working Papers I–III score a sentence as manipulative or plain. A defensive audit tool needs more than a score: it must say which rhetorical device is present, so a human reader can see the mechanism — "this is a false dilemma", "this is a loaded rhetorical question". This report documents the first steps of that explanation layer, built and tested on the 60-pair contrastive benchmark, together with the errors found and a targeted fix. Because the benchmark has only ten manipulative examples per technique — far too few to fine-tune a six-way classifier without overfitting — the classifier is zero-shot: a model that already knows the technique definitions (OpenAI gpt-4o-mini) is given the public definitions and asked to name the device. The 120 members (60 manipulative, 60 control) are each judged alone; ground truth is each manipulative member's technique, and "none" for every control.

2. Version 1: naming works, but it over-flags

When a device is present, the model names it well: accuracy 0.917 (55/60) on the manipulative members. Four of six techniques are perfect — manufactured equivalence, false dilemma, whataboutism, loaded rhetorical question, all 10/10. Presupposition is weakest (0.70; three misses labelled "none", the subtlest device), and implied causation 0.80 (two confused with false dilemma, the slippery-slope pairs).

But on plain control sentences, the classifier invents a device 65% of the time.

It says "none" correctly only 21/60 times. The clearest case is manufactured equivalence: the controls keep charged words ("dictatorship", "thief") but negate the comparison, and the model keys on the words, misses the negation, and mislabels 10/10. For a tool meant to combat manipulation, this over-flagging is the critical weakness: a detector with too many false alarms becomes an instrument of unfair accusation.

3. Version 2: instructions + abstention

Version 2 adds two things: the right to answer “none” or “insufficient_evidence”, and a declared target — name only the device the author performs and endorses (a comparison that is negated, quoted and disputed, or hedged does not count). Results on the same 60 pairs:

Indicator	v1	v2
False alarms on controls	39/60 — 65%	16/60 — 26.7%
Controls correctly “none”	21/60 — 35%	25/60 — 41.7%
Abstentions on controls	0/60	19/60 — 31.7%
Positives — correct technique	55/60 — 91.7%	51/60 — 85%
Detection recall on positives	57/60 — 95%	52/60 — 86.7%
Positives labelled “none”	3/60	8/60 — 13.3%
Abstentions on positives	0/60	0/60

False alarms fall by 38 points. But detection recall loses 8 points and correct-technique loses 7. A real defensive improvement, with a real detection cost. Crucially, abstention on positives is zero, so the drop in false alarms is not blanket refusal.

The decisive observation: exact correctness does not increase. v1 has 55+21 = 76/120 exactly-correct answers; v2 has 51+25 = 76/120 — both 63.3%, with abstention not credited. v2 is not more correct: it redistributes errors, trading false accusations for abstentions and for presupposition misses. For a defensive tool an abstention is preferable to a false accusation, so v2 is still preferable — but no gain in correctness can be claimed.

4. Why — three failure mechanisms, from the texts

1. Embedded presupposition, missed (8/8). All eight misses hide the presupposition inside a noun phrase with a loaded adjective; the author seems to be describing, and the contested judgement is backgrounded. Paradoxically, v2’s stricter target (“only what the author asserts”) pushed the model to miss them, because structurally the author asserts the frame sentence, not the presupposed adjective. Examples:

> *“Residents keep protesting the council’s FAILED housing plan.” · “Journalists exposed the senator’s SECRET foreign payments.” · “The union condemned the factory’s UNFAIR dismissal of workers.”*

2. Negated comparison, still flagged (6/6). All manufactured-equivalence false alarms are explicitly negated comparisons; the model labels them despite the negation and despite the instruction to ignore it. The general instruction did not work here. Examples:

> *“The new data law lets the state watch citizens, UNLIKE the way past dictatorships watched citizens.” · “Raising the retirement age takes money from workers, BUT NOT the way a thief takes money.”*

3. Three options read as false dilemma (8/8). All false-dilemma false alarms are three-option sentences; the model sees the enumeration frame and the binary topic but does not count the options. Examples:

> *"We could pass the security bill now, AMEND it, OR weigh the danger."* · *"The mayor could resign, apologize, OR stay while the city continues."*

These three errors are not about architecture but about compositional reasoning: negation scope, option counting, embedded presupposition. A detect-then-explain gate would not fix them by itself, since the front detector would meet the same negated comparisons and the same three-option sentences. The right fix is targeted checks per mechanism, not another global prompt.

5. A targeted negation check: a promising local result

As the first targeted check, a second pass runs ONLY on sentences labelled manufactured equivalence and asks one focused question — is the comparison negated or rejected? If yes, override to "none". Because it touches only comparison-flagged items, it cannot change any other technique. Holding the v2 predictions fixed and applying only this check (the attributable comparison):

Indicator	before	after
Equivalence false alarms (controls)	6/60	0/60
Genuine equivalence positives caught	10/10	9/10
Predictions changed outside equivalence	—	0

The dangerous negated-comparison false alarms are eliminated (6→0) at the cost of one genuine positive (Q07, where the sub-question misread an affirmed comparison as negated). Three honest limits: "zero false alarms" on these controls does not guarantee the same on new texts; Q07 is an adverse effect of the filter itself, so "no collateral damage" is restricted to the other classes; and the result supports negation-checking in this configuration, not the whole targeted-check strategy. (A methodological note: the attributable numbers hold the classifier's predictions fixed; an earlier run that re-classified with a shorter prompt gave a different baseline of 9, which is not comparable.)

6. The next check: false dilemma is not "exactly two options"

The next sieve must not count options. Three options can still be a falsely closed choice, and two can be perfectly legitimate:

Two, legitimate: "You can file the application online or at the counter; both are available."

Three, falsely exhaustive: "Either you support the project, or you are incompetent, or you have a hidden agenda."

The verifier's question should target the logical mechanism: "Does the text present the alternatives as the only possible ones and unjustifiably exclude other relevant possibilities?" With three outcomes — explicitly open alternatives: reject the label; unjustified closure supported by the text: keep it; undeterminable: abstain. The option count is a descriptive hint, not the decision rule.

7. Roadmap

In order: document and save the current version (done); fix Q07 by narrowing the question to the negation of the equivalence relation rather than any negation; build the verifier for unjustified exhaustiveness of alternatives; test it on legitimate two-option choices and falsely-closed three-option ones; and validate on an untouched set after tuning. A methodological rule going forward: a verifier should load saved predictions rather than re-run the classifier, so its effect stays attributable. The broad lesson so far is modest and real — targeted, per-mechanism checks are a promising path, and one of them (negation) works cleanly on this run; the strategy as a whole is not yet validated.

Materials

All scripts and prediction files are in the repository: the technique classifier (13_technique_classifier.py), its metrics (14_technique_metrics.py), the v2 abstention classifier (16_classifier_v2_abstention.py), the hard-control set (hard_controls.csv), the targeted negation check (17_negation_check.py), and the per-sentence predictions for every run.